

The Normal Distribution

For a random variable X , the Z - score (also known as the standard score) tell sus how many standard deviations X is above or below the mean.

$$z = \frac{x - \mu}{\sigma}$$

Z scores allow for comparisons between variables from different normal distributions.

An observation x_1 is said to be more unusual (or more of an outlier) than another observation x_2 if $z_1 > z_2$.

The standard deviation of a normal distribution is

$$\sqrt{np(1 - p)} \quad (1)$$

and mean is np .

1 _Definition_: Percentile

The percentile of a is the percentage of outcomes from a distribution that are lower than a . I.e. $P(x < a) = \int_{-\infty}^a f(x)dx$ where $f(x)$ is the probability distribution.

Lets do a practice problem. 11.5% of the population smokes. 400 randomly selected people are surveyed and 42/400 are asked. What is the probability of seeing 42 or less?

The bad way of doing this is binomial:

$$P(\# \leq 42) = P(x = 0) + P(x = 1) + \dots + P(x = 42) \quad (2)$$

$$= \sum_{i=0}^{42} 10.115^i * (1 - 0.115)^{400-i} \quad (3)$$

The better way: We can say that the mean is $400 * 0.115$, $\sigma = \sqrt{400 * 0.115 * (1 - 0.115)}$

We can get the z score as

$$z = \frac{(42) - \mu}{\sigma} \quad (4)$$

$$\text{so we have} \quad (5)$$

$$p(z \leq 0.063) = 0.7357 \quad (6)$$

You can calculate the z score, then do `normcdf(-99,z,0,1)` since the z scores follow a normal distribution with $\mu = 0$ $\sigma = 1$.